GenAI Peer Learning Companion Integration & API Guide

Technical Annex

1. Platform Overview

The service is an OpenAI-compatible inference API running open-source language models on CPU-only, GPU-free cloud infrastructure. Each team receives one dedicated, non-transferable API key. Prompts, models and vector indexes remain on this infrastructure for the duration of the challenge.

2. Model Classes

Class	Size	Capacity
Compact	< 3B params	≈ 60%
Mid-size	≤ 16B params	≈ 40%

- **Compact:** classification, intent detection, extraction, formatting, rewriting, short summaries.
- **Mid-size:** reasoning, multi-source synthesis, planning, final user-facing writing.
- The 60/40 split is what allows several teams to be served at once without GPUs, and what makes the exercise meaningful: the scarce resource is not "the model", it is compute time on the mid-size class.

3. API Access

- **Base URL:** communicated before the challenge opens. Must be a configurable value, never hard-coded.
- **Endpoints:** /v1/models, /v1/chat/completions, and a separately metered embeddings endpoint.
- **Auth:** Authorization: Bearer <team_api_key> header on every request. Quota and metrics are attached to the key.
- **SDK compatibility:** OpenAI Python/JS SDK, LangChain, LlamaIndex, or a plain HTTP client.
- Responses are returned whole (no token streaming).

4. Retrieval (RAG) Infrastructure

- A vector database is deployed by the platform, with one dedicated collection per team.
- The embeddings endpoint has its own quota, separate from the generation quota.

5. Quota Mechanism

Quota works as a token bucket: each request consumes one unit, and the bucket refills continuously at a set rate, up to a ceiling. Refilling is continuous, so there is no reset at a fixed clock boundary.

- **Sustained rate:** the durable working pace of the key.
- **Burst reserve:** allows an occasional salvo (e.g. indexing a batch of documents) before returning to the normal regime.
- **Additional ceilings:** a maximum response length per request, and a maximum number of concurrent requests per key.
- The embeddings quota is separate from the generation quota

Tiers: Three tiers apply to the same key over the challenge lifecycle:
Teams are encouraged to go beyond a simple chat

Phase	Tier
Integration (pre-event) Challenge (during event) Demonstration (jury slot)	Reduced Baseline Widened - no team is quota limited

6. Response & Error Contract

Situation	Service Response	Expected Handling
Team quota exhausted	Team quota exhausted	Wait the stated delay, then retry. The service supplies the delay.
Model class saturated	503 + Retry-after	Retry at intervals, or fall back to the compact class.
Request beyond a ceiling	400	Fix the request; the error message identifies the ceiling exceeded
Normal operation	200 + quota headers	Read the remaining quota and pace requests accordingly

This contract is stable for the whole challenge, so resilience logic can be built and tested against it in advance.

7. Orchestration Guidance

Step	Step
Classification, triage, intent detection	Compact
Field extraction, formatting, rewriting	Compact
Short summary of targeted text	Compact
Reasoning, synthesis, planning	Mid-size
Final user-facing writing	Mid-size
Fallback when mid-size saturated	Compact

Useful patterns: escalation (compact handles the request, mid-size only when needed), draft-then-revise (compact drafts, mid-size corrects), plan-once/execute-compact (one mid-size call plans, compact executes each step), and narrow-before-generating (vector search + compact filter shrink context before the mid-size call).

8. Per-Team Metrics

- Calls made — total and per model.
- Compact / mid-size split — the orchestration indicator.
- Tokens consumed — input and output.
- Repeated identical requests — the cache indicator.
- Refusals on quota and how they were handled.

Metrics are shared back with each team during the event, limited to its own key.

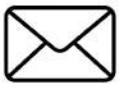
9. Evaluation Weighting

Criterion	Weight
Model orchestration	25%
Handling the constraints	25%
Relevance & usefulness	20%
Resilience under load	15%
Originality	10%
Presentation	5%

Proposed weighting for the platform-related portion of scoring, published by the organising team before the challenge opens, alongside the general grid (Innovation, Technical Feasibility, Relevance, Impact, Pitch & Demo) defined in the specification book.

10. Usage & Security Rules

- One key per team, non-transferable. No sharing or exposure in code, logs, screenshots or videos.
- No bypassing quota via extra keys, accounts, or a replacement API during the demo. Declare any additional external AI service.
- Only public, fictional, synthetic, or anonymised data.
- Technical call metadata (timestamp, model, token volume, return code) is logged for load observation. Logs are purged 30 days after the event.
- Teams retain full ownership of their code.
- A technical support channel is available for persistent service errors expected latency and queueing do not fall under it.



For any inquiries or clarifications, please contact us at wie.act@ieee.tn